Two Elementary Inequalities

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Abstract

We prove the two-variable inequality between the arithmetic and the geometric mean, and a cubic inequality in three variables that follows from Schur's inequality.

1 Preliminaries

We use one classical result, which we cite without proof.

Lemma 1 (Schur's inequality [1, Theorem 1])**.** *For all real numbers $a, b, c \geq 0$,*

$$a(a-b)(a-c) + b(b-a)(b-c) + c(c-a)(c-b) \geq 0.$$

Lemma 2. *For all real numbers x and y, $x^2 + y^2 \geq 2xy$.*

Proof. $x^2 + y^2 - 2xy = (x-y)^2 \geq 0$. □

2 Main results

Theorem 3. ***Original:***
For all real numbers a and b,

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

Proposed repair (awaiting your decision):
For all real numbers $a, b \geq 0$,

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

TheoremChecker:
The proposed repair restates the result as in blue. False for negative numbers: for $a = b = -1$ the left side is -1 while the right side is $\sqrt{1} = 1$. The proof takes $\sqrt{a}$ and $\sqrt{b}$, which are real numbers only when $a, b \geq 0$. Counterexample: $a = b = -1$: $\frac{a+b}{2} = -1 < 1 = \sqrt{ab}$.
Gives up: The inequality is no longer claimed when a or b is negative.

*A made-up paper for the TheoremChecker demo.

Proof. Apply Lemma 2 to $x = \sqrt{a}$ and $y = \sqrt{b}$: then $x^2 = a$, $y^2 = b$ and $xy = \sqrt{ab}$, so $a + b \geq 2\sqrt{ab}$. □

Proof of the proposed statement:
Apply Lemma 2 to $x = \sqrt{a}$ and $y = \sqrt{b}$, real numbers since $a, b \geq 0$: then $x^2 = a$, $y^2 = b$ and $xy = \sqrt{ab}$, so $a + b \geq 2\sqrt{ab}$.

Theorem 4. *For all real numbers $a, b, c \geq 0$,*

$$(a+b+c)^3 + 9abc \geq 4(a+b+c)(ab+bc+ca).$$

Proof. Expanding both sides,

$$(a+b+c)^3 + 9abc - 4(a+b+c)(ab+bc+ca) = a^3 + b^3 + c^3 - \sum_{\text{sym}} a^2 b + 3abc,$$

where $\sum_{\text{sym}} a^2 b$ is the sum of the six terms $a^2 b$, $a^2 c$, $b^2 a$, $b^2 c$, $c^2 a$, $c^2 b$. The right-hand side equals $a(a-b)(a-c) + b(b-a)(b-c) + c(c-a)(c-b)$, which is nonnegative by Lemma 1. □

References

[1] G. H. Hardy, J. E. Littlewood and G. Pólya, *Inequalities*, 2nd ed., Cambridge University Press, 1952.